THANGAM G

Biomedical Engineering Student | Medical Device R&D | Product Development

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PROFESSIONAL SUMMARY

Biomedical Engineering student and aspiring Product Engineer skilled in CAD/CAM, embedded systems, PCB design, and medical device development. Driven to transform innovative ideas into impactful healthcare products through design, prototyping, and engineering, with a focus on medical devices and healthcare robotics.

SKILLS

TECHNICAL

- **Design & Manufacturing:** Fusion 360,Solidworks (CAD/CAM), 3D Modeling, Assembly Design, Engineering Drawings, DFM, Prototyping
- **Embedded Systems & Electronics:** Arduino, STM32, Embedded C, Sensor Integration, Circuit Analysis, Hardware Interfacing
- **IoT & Healthcare Technologies:** Medical Devices, IoT Healthcare, Wearable Devices, Biomedical Instrumentation, Product Lifecycle Management (PLM), Design Controls, Verification & Validation (V&V), Risk Management, Quality Assurance, Regulatory Compliance, ISO 13485, 14971
- **Simulation & Modeling:** MATLAB, System Simulation, Control Systems
- **PCB Design:** KiCad, Altium Designer, PCB Layout & Routing, PCB Design Fundamentals
- **Programming & AI s:** Python, C/C++, Java, AI/ML, Generative AI, Computer Vision, Data Analysis.

SOFT

- **Communication:** Presentation, Documentation, Public Speaking.
- **Problem Solving:** Critical Thinking, Decision Making, Attention to Detail.
- **Leadership & Teamwork:** Collaboration, Leadership, Coordination.
- **Innovation:** Design Thinking, Creativity, Product Innovation.
- **Professional Skills:** Adaptability, Time Management, Continuous Learning.

EDUCATION

B.E. Biomedical Engineering [CGPA : 8.1]
SNS College of Technology, Coimbatore | 2023–2027
Higher Secondary Certificate (HSC)
M.N.R.D Girls Higher Secondary School, Srivilliputhur | 2021–2023
Secondary School Leaving Certificate (SSLC)
M.N.R.D Girls Higher Secondary School, Srivilliputhur | 2019–2020|

PROJECTS & INTERNSHIPS

PROJECTS

- **3D Bioprinting of Sacrum | 2023 -** Developed a bioprinted sacrum model to support anatomical reconstruction, surgical planning, and tissue engineering research.
- **LED-Based Blood Glucometer | 2024 -** Developed a prototype using Arduino and LED optical sensing, reducing component cost by ~30% compared to conventional glucometer designs.
- **Catalyst Synthesis from Spent Lithium-Ion Batteries | 2025 -** Recovered and repurposed lithium-ion battery cathode materials to develop catalysts for sustainable hydrogen production.
- **Intelligent Pesticide Sprinkling System | 2025 -** Designed an IoT-based automated spraying system that reduced pesticide wastage and enabled remote monitoring of field operations.
- **Industrial Automation Bot (Ongoing) | 2026–Present -** Developing an embedded automation system integrating sensors and intelligent control mechanisms to enhance process efficiency and real-time monitoring.

INTERNSHIPS

- **InnAccel Technologies Pvt. Ltd., Bangalore | Dec 2025 – Jan 2026 -** Medical Device Development, Product Design, Prototyping , Innovation-Driven R&D
- **National Institute of Technology (NIT), Trichy | Jun 2025 – Jul 2025 -** Assisted in Mechanical Engineering research involving CAD, simulation, and data analysis.
- **Mindful-AI, Coimbatore | Jul 2024 – Oct 2024 -** AI-Driven Healthcare Solutions, Machine Learning, and Analytics
- **Johnson & Johnson – Robotics & Controls (Virtual) | May 2026 -** Robotics, Control Systems, and Healthcare Automation
- **Naachiaar Bandage Mill, Rajapalayam | Feb 2025 -** Healthcare Manufacturing, Quality Assurance, and Sterilization Processes
- **Karthick Ortho Hospital, Rajapalayam | May 2024 -** Orthopedic Clinical Workflows, Biomedical Equipment, and Rehabilitation Technologies

CERTIFICATIONS & ACHIEVEMENTS

CERTIFICATIONS

- Mechanical Product Design using CAD/CAM – Autodesk
- STM32 Microcontroller Programming & Embedded Systems – Great Learning
- Electronics & Sensor Integration – LinkedIn Learning , Electrical Measuring Instrumentation – Alison
- PCB Design & Layout Engineering (KiCad & Altium Designer) – Simplilearn & Altium
- ISO 13485,ISO 14971-SkillMed
- Industrial Automation & PLC Programming,IoT Device Management & Operating Systems – LinkedIn Learning
- Lifelong Professional Skills - IBM , AI Fundamentals: Foundations for Understanding AI – IBM SkillsBuild

ACHIEVEMENTS

- Best Poster Award – IoT-Based Heart & Respiratory Monitoring System (2024)
- 4th Place – Hack-Cure Hackathon, VIT Chennai (2025)
- 3rd Place – Igninium Technical Competition (2026)
- Participated in 10+ Hackathons and Technical Competitions